

Library Fresh
main course
ML, DL, NLP

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lec - 2 S

Understanding Complete
Linear Regression is
Constructive

lec - 4 Convergence
Algorithm
Part 1

→

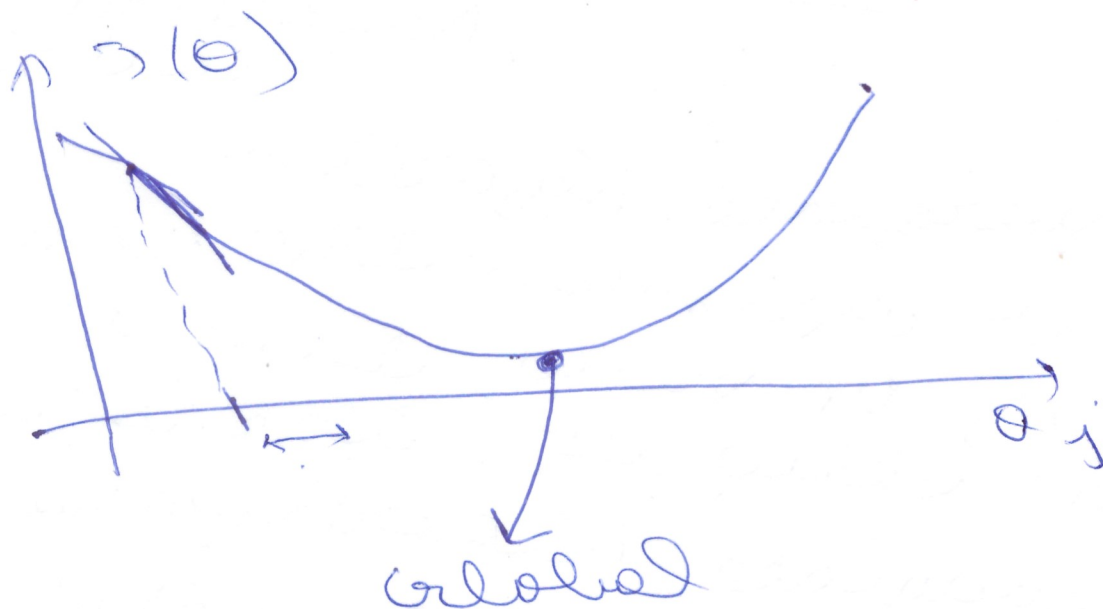
lec - 5 Convergence
Algorithm
Part - 2

⇒ Convergence Algorithm
, it optimizes the change
of 0, value, that is
the value of the slope
Convergence Algorithm basically
says that repeat until
convergence, that until
we reach the global
minimum point.

• If we reach near the
global minimum, then
we get the best fit
line

⇒ So basically, we repeat
until convergence..

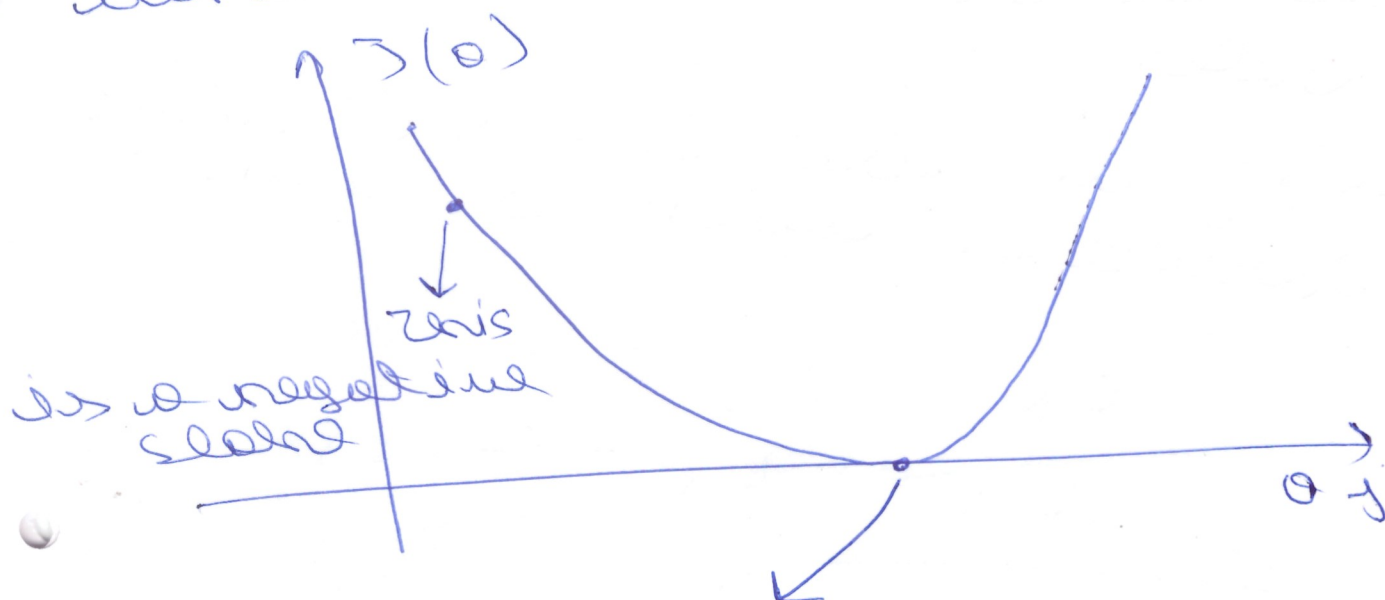
$$\theta_j = \theta_j - \alpha \left[\frac{\partial}{\partial \theta_j} J(\theta_j) \right]$$



• Here, basically, we're
trying to calculate the
slope

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of the need to understand whether it's a peak or not



It basically means that whenever we try to find a derivative

$$\theta_j = \theta_j - \underbrace{\frac{\partial}{\partial \theta_j} f(\theta_j)}_{\text{negative}}$$

This value will be negative

Suppose we get a negative value, then

$$\theta_j = \theta_j - \alpha$$

$$\theta_j^- = \theta_j + (ve)$$

→ Here in θ_j , basically we're adding some +ve values

→ If we do this iterative process again & again -

$$\theta_j = \theta_j - \alpha(-ve)$$

$$\theta_j = \theta_j + (+ve)$$

$$\frac{\partial}{\partial \theta_j} J(\theta_j)$$

It can be represented as -

$$\begin{aligned} \theta_j^- &= \theta_j - \alpha(+ve) \\ &= \theta_j - (+ve) \end{aligned}$$

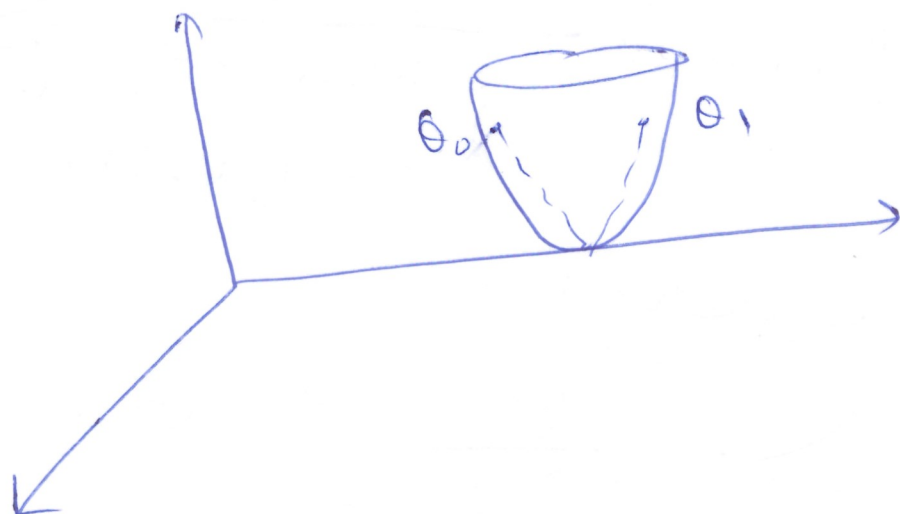
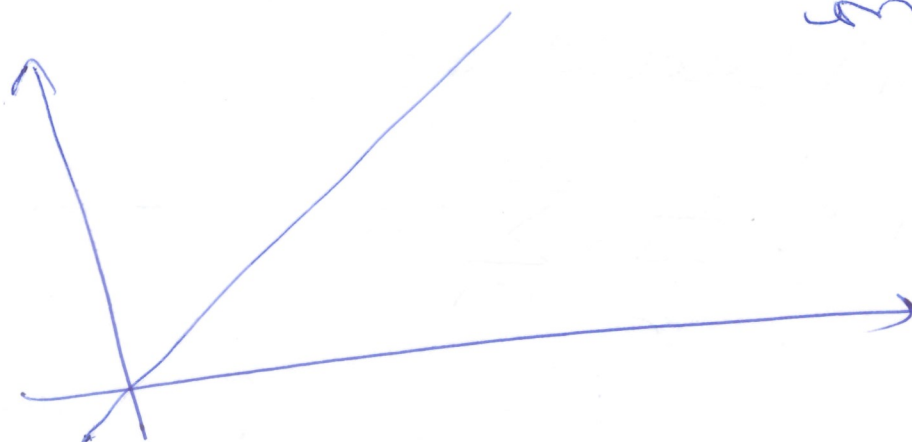
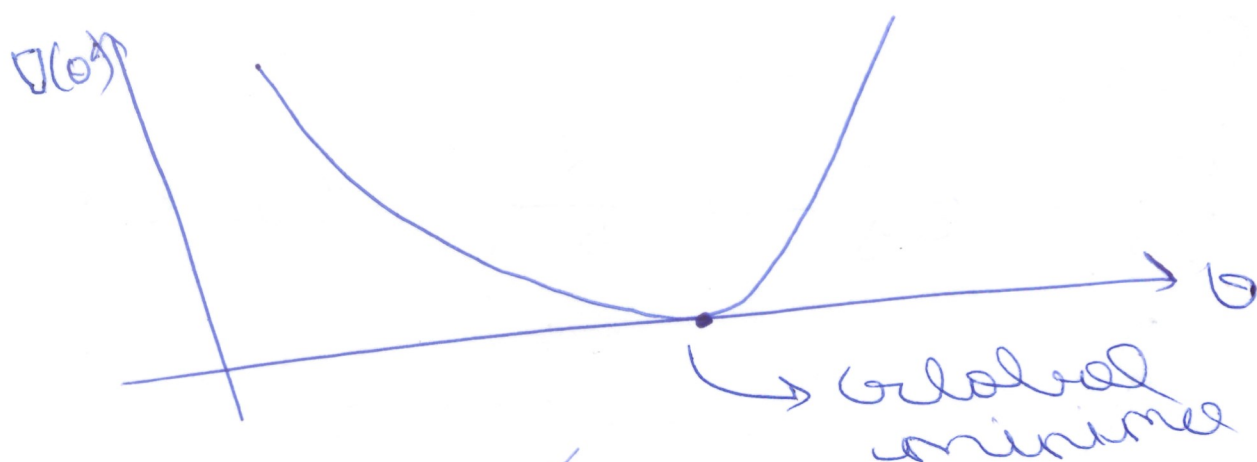
$$\alpha = \text{Learning Rate} \\ \alpha = 0.001$$

→ This process continues until we don't come near the global minimum

(5)

Final Conclusion

Gradient descent :-



Convergence Algorithm

→ we repeat the convergence algorithm, until it reaches ϵ :-

$$\left\{ \begin{array}{l} \theta_j = \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) \end{array} \right.$$

∴, $J(\theta_0, \theta_1)$ will be

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^i) - y^i)^2$$

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) = \frac{\partial}{\partial \theta_j} \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^i) - y^i)^2$$

$y = 0$ and 1

$$\rightarrow \frac{\partial}{\partial x} x^4 = 4x^3 - 1$$

$$\frac{\partial}{\partial x} (x)^2 = 2x$$

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) = \frac{\partial}{\partial \theta_j} \times \frac{1}{2m} \sum_{i=1}^m (\theta_0(x)^i - y^{(i)})^2$$

der

$$J = 0 \Rightarrow \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{\partial}{\partial \theta_0} \times \frac{1}{2m} \left[\sum_{i=1}^m (\theta_0(x)^i - y^{(i)})^2 \right]$$

$$J = 0 \Rightarrow \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{\partial}{\partial \theta_0} \times \frac{1}{2m} \sum_{i=1}^m (\theta_0(x)^i - y^{(i)})^2$$

//

the entire value will get converted to:

$$= \frac{1}{m} \sum_{i=1}^m (\theta_0(x)^i - y^{(i)})^2$$

$$J = 1 \Rightarrow \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = \frac{\partial}{\partial \theta_1} \times \frac{1}{2m} \left[\sum_{i=1}^m (\theta_0 + \theta_1 x) - y^{(i)} \right]^2$$

$$= \frac{1}{m} \sum_{i=1}^m ([\theta_0 + \theta_1 x] - y^{(i)})^2$$

$$\frac{\partial}{\partial \theta_i} [\theta_0 + \theta_1 x] = 1 \text{ or } x$$

(1) we repeat until the convergence theorem is

$$\theta_0 := \theta_0 - \alpha \left(\frac{1}{m} \sum_{i=1}^m (\theta_0(x)^{(i)} - y^{(i)}) \right)$$

$$\theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m (x(x)^{(i)} - y^{(i)})$$

so $\vec{\theta} = \begin{pmatrix} \theta_0 \\ \theta_1 \end{pmatrix} \Rightarrow$

$$\frac{\partial}{\partial \theta_i} J(\theta_0, \theta_1)$$

$$= \frac{\partial}{\partial \theta_i} \frac{1}{2m} \left[\sum_{i=1}^m (\theta_0 + \theta_1 x - y^{(i)})^2 \right]$$

$$\frac{\partial}{\partial \theta_i} [\theta_0 + \theta_1 x] = 1 \text{ or } x$$

Here, by these equations we get the best fit line

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$$\frac{\partial}{\partial \theta_1} [\theta_0 + \theta_1 x] = 1$$

we repeat with convergence

$$\theta_0 = \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (\theta_0(x_i) - y_i)$$

$$\theta_1 = \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m (x_i(\theta_0(x_i) - y_i))$$



Here, we repeat it until convergence